

Programming Challenges

1. Scrollable Tax Calculator

Build a JavaFX application that lets the user enter a purchase amount into a `TextField` and displays the resulting sales tax. Use a `Slider` to let the user adjust the tax rate anywhere between 0 percent and 10 percent, updating the displayed tax as the slider moves.

2. Image Viewer

Build a JavaFX application that lets the user browse and view image files. Provide a `Button` or menu item that opens a `FileChooser` dialog; once the user selects an image file, load it and display it in an `ImageView`.

3. Dorm and Meal Plan Calculator

A university offers the following dormitories: Allen Hall — \$1,800/semester; Pike Hall — \$2,200/semester; Farthing Hall — \$2,800/semester; University Suites — \$3,000/semester. It also offers these meal plans: 7 meals/week — \$600/semester; 14 meals/week — \$1,100/semester; unlimited meals — \$1,800/semester. Build a JavaFX application with two `ComboBox` controls — one listing dormitories, the other meal plans — that displays the combined semester charge once both selections are made.

4. Skateboard Designer

The Skate Shop sells the following skateboard components.

Decks

	Truck Assemblies	Wheels
The Master Thrasher — \$60	7.75-inch axle — \$35	51 mm — \$20
The Dictator — \$45	8-inch axle — \$40	55 mm — \$22
The Street King — \$50	8.5-inch axle — \$45	58 mm — \$24
		61 mm — \$28

The shop also sells grip tape (\$10), bearings (\$30), riser pads (\$2), and a nuts & bolts kit (\$3). Build a JavaFX application that lets the user pick one deck, one truck assembly, and one wheel set using `ListView` or `ComboBox` controls, plus a `ListView` that

allows selecting multiple miscellaneous items. Display the subtotal, 7 percent sales tax, and order total.

5. Shopping Cart System

Build a JavaFX shopping-cart application for an online bookstore. At start-up, read a `BookPrices.txt` file containing comma-separated title/price pairs (one book per line) and load the titles into a `ListView`. Let the user select a title and add it to a second `ListView` representing the shopping cart. Provide controls to remove an item from the cart, clear the cart, and check out; checking out should display the cart subtotal, 7 percent sales tax, and the total.

6. Smartphone Packages

Cell Solutions sells the following monthly data plans: 8 GB — \$45.00; 16 GB — \$65.00; 20 GB — \$99.00. It also sells these phones (a 6 percent sales tax applies to phone purchases): Model 100 — \$299.95; Model 110 — \$399.95; Model 200 — \$499.95. Customers may also add Phone Replacement Insurance (\$5.00/month) and WiFi Hotspot Capability (\$10.00/month). Build a JavaFX application with a Menu-based interface that lets the user pick one data plan, one phone, and any combination of add-ons, updating the displayed price as selections change.

7. Shade Designer

A custom window-shade designer charges a \$50 base fee per shade, plus style, size, and color add-ons.

- **Styles:** Regular +\$0, Folding +\$10, Roman +\$15
- **Sizes:** 25" +\$0, 27" +\$2, 32" +\$4, 40" +\$6
- **Colors:** Natural +\$5, Blue/Teal/Red/Green +\$0

Build a JavaFX application that lets the user choose the style, size, color, and quantity of shades from `ListView` or `ComboBox` controls, and displays the total charge.

8. Conference Registration System

Build a JavaFX application that calculates conference registration fees. General registration is \$895/person and student registration is \$495/person. An optional opening-night dinner with a keynote speech is \$30/person. The following preconference workshops are also available, and any number may be selected via `CheckBox` controls.

Workshop	Fee
Introduction to E-commerce	\$295
The Future of the Web	\$295

Advanced Java Programming	\$395
Network Security	\$395

Let the user select the registration type, opt into the dinner/keynote, and check any number of workshops, then display the total cost.

9. Long Distance Calls

A long-distance provider charges by time of day: Daytime (6:00 a.m.–5:59 p.m.) — \$0.07/minute; Evening (6:00 p.m.–11:59 p.m.) — \$0.12/minute; Off-Peak (12:00 a.m.–5:59 a.m.) — \$0.05/minute. Build a JavaFX application with a group of `RadioButton` controls (in a shared `ToggleGroup`) for selecting the rate category, plus a `TextField` for the call length in minutes. Display the resulting charge in an `Alert` dialog.